

TASK 2 - REPORT

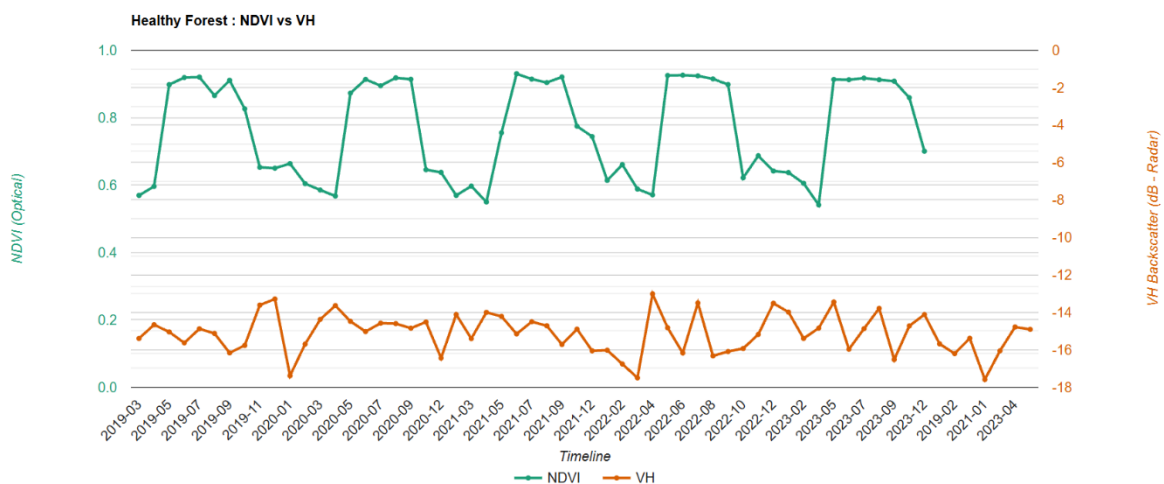
1. Selected Data Points

The following 6 loss pixels within the Bavaria bounding box were selected and analysed across a multi-year window (2019–2023):

Pixels	Category	Tree Cover (%)	Loss Year	Coordinates
Pixel 1	Healthy Forest	-	-	(11.73603, 47.9334)
Pixel 2	Dense 1	98	2022	(11.4376, 48.0569)
Pixel 3	Dense 2	89	2020	(11.62360, 47.98457)
Pixel 4	Medium	64	2022	(11.29928, 48.04259)
Pixel 5	Sparse	47	2020	(11.448847, 48.015903)
Pixel 6	Very Sparse	21	2020	(11.604945, 47.873129)

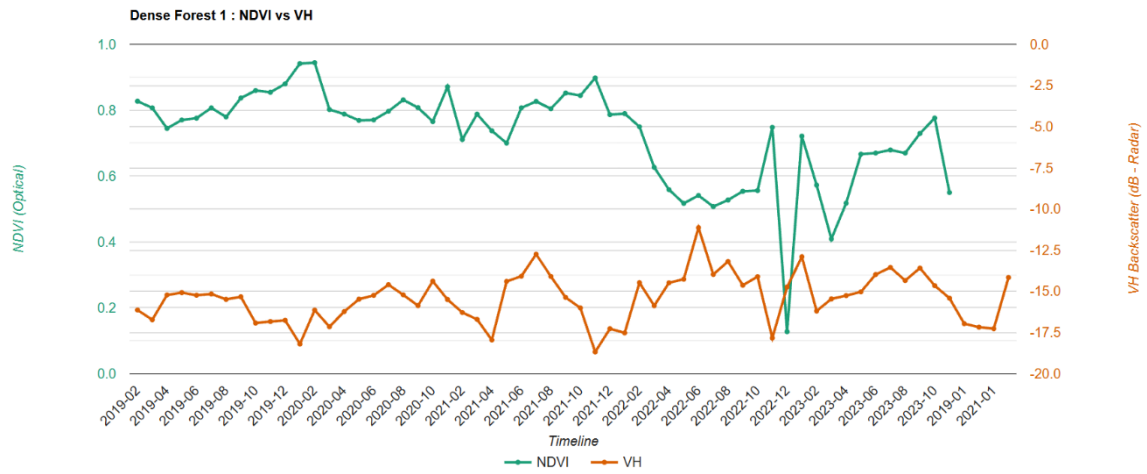
2. NDVI vs VH Time-Series Charts:

A) Healthy Forest Pixel -



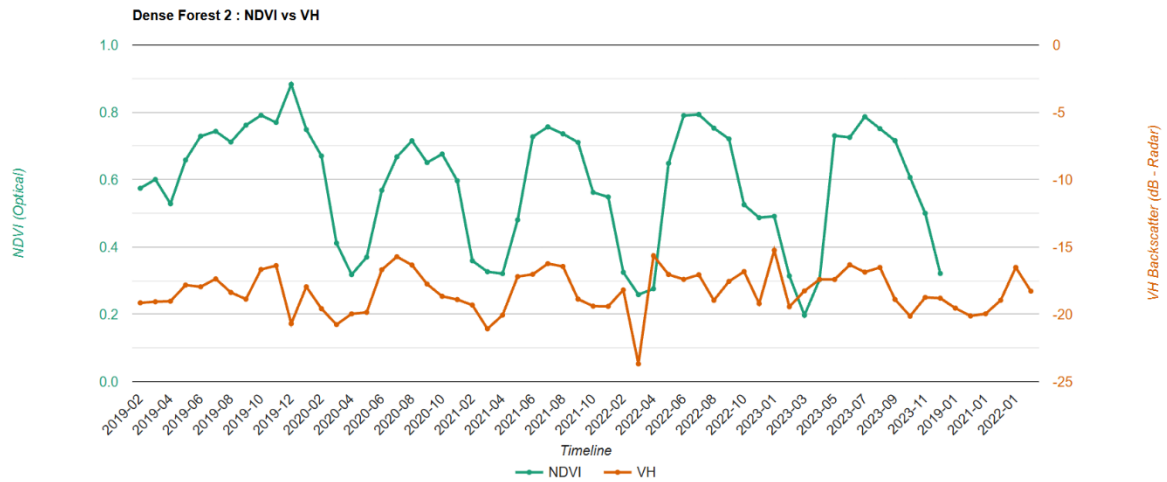
Healthy Forest Pixel NDVI vs VH Time Series

B) Dense Forest Pixel (1) -



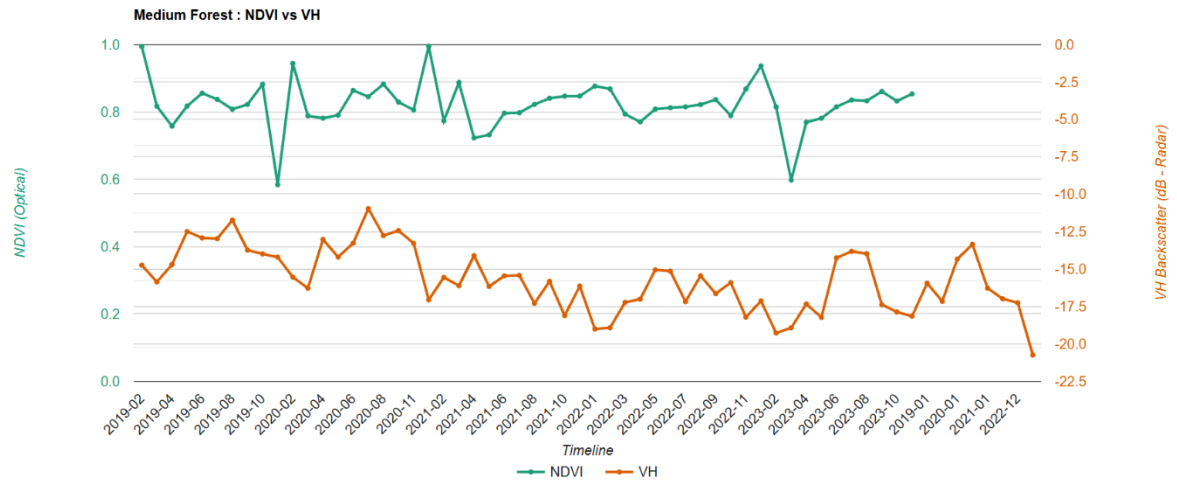
Dense Forest Pixel (1) NDVI vs VH Time Series

C) Dense Forest Pixel (2) -



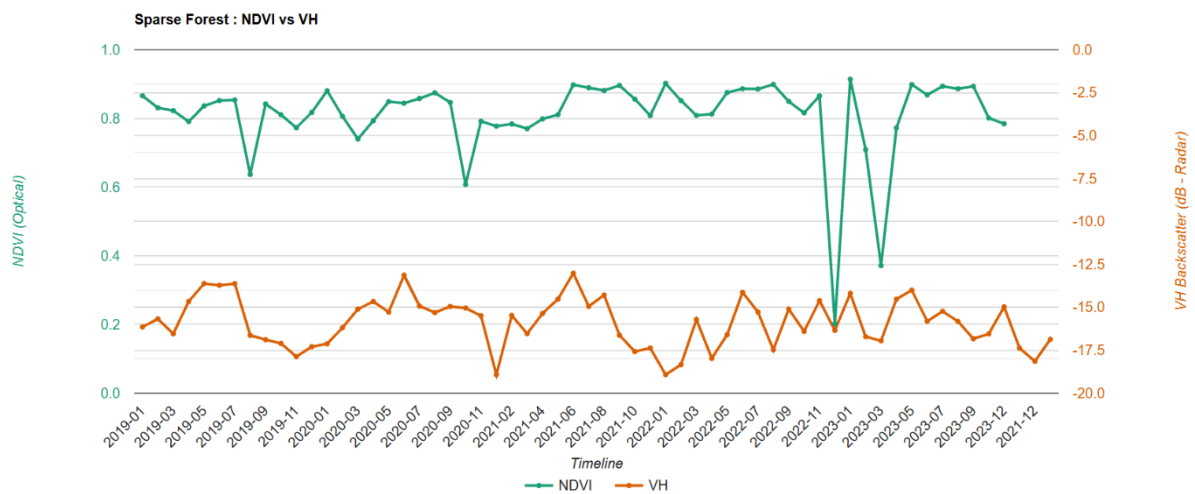
Dense Forest Pixel (2) NDVI vs VH Time Series

D) Medium Forest Pixel -



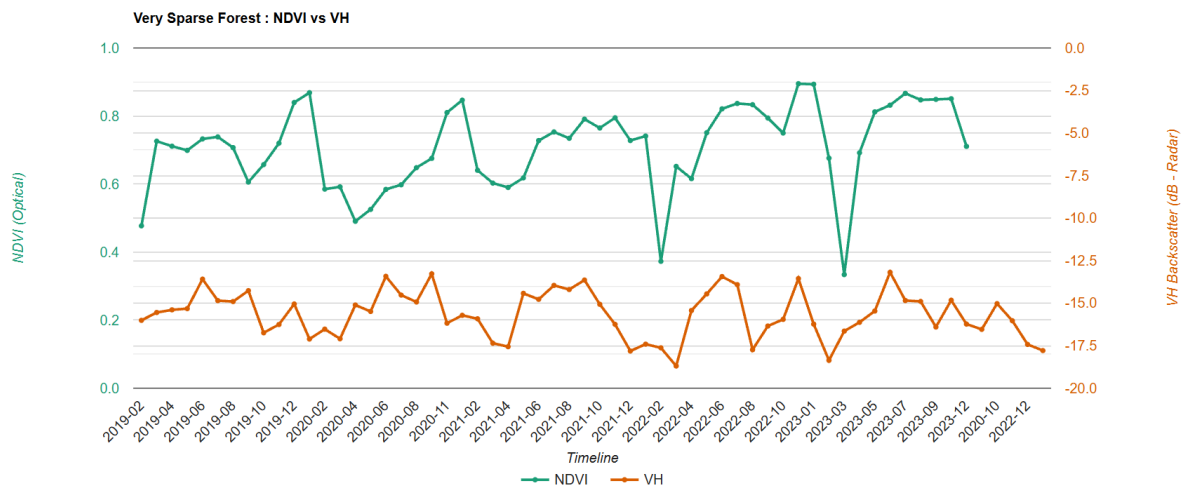
Medium Forest Pixel NDVI vs VH Time Series

E) Sparse Forest Pixel -



Sparse Forest Pixel NDVI vs VH Time Series

F) Very Sparse Forest Pixel -



Very Sparse Forest Pixel NDVI vs VH Time Series

Verdicts per Pixel:

1) Pixel 1 (Healthy Forest):

Observation:

NDVI exhibits a clear and repeatable seasonal cycle, with summer values around **0.89 - 0.93** and winter values around **0.54 - 0.70**. The forest consistently recovers to similar summer NDVI values each year, indicating stable vegetation throughout the study period. VH backscatter remains comparatively stable, fluctuating approximately between **-13 dB and -17.5 dB** without a recurring seasonal decline comparable to NDVI.

*For the stable forest, NDVI exhibits strong seasonal variability whereas VH remains relatively stable. This suggests that normal leaf-off alone does not produce a sustained VH decrease.

2) Pixel 2 (Dense Forest 1): [LOSS YEAR: 2022]

Observation:

Prior to 2022, NDVI remained consistently high (approximately **0.75–0.94**), indicating a dense and stable forest canopy. Around the confirmed Hansen loss year (2022), NDVI declined substantially and remained suppressed throughout the growing season rather than recovering to previous levels. VH responded differently from NDVI and became generally less negative after the disturbance.

*The VH values didn't reduce post loss year. This suggests radar waves bounce off this chaotic, rough debris field intensely, causing high radar backscatter volume despite the loss of green canopy.

3) Pixel 3 (Dense Forest 2): [LOSS YEAR: 2020]

Observation:

The 2020 disturbance is clearly visible in the NDVI series and is accompanied by an equally distinct change in VH backscatter. Around the Hansen loss year (2020), NDVI drops sharply from approximately 0.79 to 0.32, while VH simultaneously becomes more negative (around -20 dB), indicating a reduction in vegetation structure alongside the loss of green biomass. VH is observed to be noisier than NDVI throughout the record, the combined NDVI–VH response provides stronger evidence of a genuine forest disturbance than NDVI alone.

* VH continues to fluctuate seasonally, but the disturbance period is clearly identifiable through the coincident decline in NDVI and VH.

4) Pixel 4 (Medium): [LOSS YEAR: 2022]

Observation:

NDVI remained consistently high throughout the 2022 growing season, showing no obvious reduction in canopy greenness around the mapped loss year. In contrast, Sentinel-1 VH became more negative during late 2021 and early 2022, suggesting changes in forest structure before any noticeable change appeared in the optical signal.

* However, VH values also exhibit considerable natural variability throughout the record, making it difficult to attribute this change solely to forest loss.

5) Pixel 5 (Sparse): [LOSS YEAR: 2020]

Observation:

NDVI remained consistently high throughout the reported loss year, with no sustained reduction in vegetation greenness. Sentinel-1 VH backscatter remained within its historical range throughout the entire study period and did not exhibit a distinct structural change during either the reported loss year or the later NDVI declines.

* VH did not distinguish a real clearing event for Pixel 5. Neither the Hansen loss year (2020) nor the later NDVI declines were accompanied by a sustained decrease in VH backscatter. Short lived anomalous drops with instant recovery were masked as outliers.

6) Pixel 6 (Very Sparse): [LOSS YEAR: 2020]

Observation:

NDVI exhibited temporary reductions during early 2022 and early 2023; however, these declines recovered rapidly and did not coincide with the mapped Hansen loss year. Sentinel-1 VH backscatter remained within its historical range throughout the study period and showed no sustained decrease corresponding to either the reported loss year or the later NDVI reductions. No visible synchronisation of optical and radar changes is observed.

* VH did not successfully distinguish a forest-clearing event for Pixel 6.

Overall, Sentinel-1 VH provided useful structural information, but it did not consistently distinguish normal seasonal leaf-off from real forest clearing across all pixels.

VH successfully distinguished leaf-off from real clearing only in some dense forest pixels overall, its behaviour was noisier and less consistent than expected, making it a complementary rather than standalone indicator of forest disturbance.

GEE Code Link: <https://code.earthengine.google.com/b4534b87e7a2abd63714080151bf60be>

GitHub Repository Link: <https://github.com/SiddharthBhimpure/bavaria-forest-change-analysis.git>

END OF REPORT